Coastal Ecology Lab, Puget Sound Institute of Science

Standard Operating Procedure

SOP CE-04: Rocky Intertidal Quadrat Sampling

Document:	SOP CE-04, version 2.1
Approved:	September 3, 2025
Prepared by:	Yolanda Reeves, Lab Manager
Approved by:	Dr. Sarah Lindstrom, Principal Investigator
Applies to:	All field personnel collecting intertidal community data

1. Purpose

To standardize the collection of invertebrate abundance and percent-cover data along fixed intertidal transects so results are comparable across sites and dates.

2. Equipment

- 0.25 m by 0.25 m PVC quadrat with string grid (25 intersections)
- 50 m fiberglass transect tape and stakes
- Waterproof field tablet or Rite in the Rain notebook and data sheet
- Handheld thermometer (water and air), GPS unit
- Camera for quadrat photographs, scale bar
- Field guide (Between Pacific Tides) for in-field confirmation

3. Procedure

1. Arrive at the site at least 60 minutes before predicted low tide. Confirm the tide height against the NOAA Tacoma station.
2. Lay the transect perpendicular to the waterline from the high to low intertidal zone. Record start GPS coordinates.
3. Place quadrats at fixed 5 m intervals along the transect.
4. Record water and air temperature at each transect start.
5. For each quadrat, count all mobile invertebrates and estimate percent cover of sessile taxa using the 25 grid intersections.
6. Photograph each quadrat with the scale bar visible.
7. Enter data the same day. Flag any temperature reading outside 5 to 25 C for re-check.

4. Data handling

Raw counts are exported as tab-separated files and archived in the shared lab Drive folder. File naming follows Intertidal_Survey_RawCounts_YEAR. Do not edit raw files; corrections go in a separate log.